ASU Micro Qual Module 6

Moral Hazard, Principal–Agent, and Mechanism Design including Auctions

First-year PhD Microeconomic Theory Qualifier Packet
Overleaf-ready source file

Contents

1 Orientation

This packet is designed for exam execution. The target skill is not to recite moral hazard or auction theory, but to recognize the object being chosen, write the exact constraints, reduce the problem with the right theorem, and verify the candidate mechanism or contract. Every canonical model below is written in the required four-layer form:

$$\textbf{Object} \quad \to \quad \textbf{Feasibility} \quad \to \quad \textbf{Optimality/Equilibrium} \quad \to \quad \textbf{Verification}.$$

Use this module after Modules 3–5. Module 3 supplies equilibrium definitions; Module 4 supplies Bayesian-game and belief discipline; Module 5 supplies IC/IR and monotone-menu logic. Module 6 turns those primitives into hidden-action contracts, direct revelation mechanisms, envelope arguments, and auction revenue formulas.

Estimated study use: two intensive blocks. Block 1: moral hazard, FOA, Grossman–Hart cost minimization, monitoring. Block 2: direct revelation, envelope, Myerson, revenue equivalence, first-price bidding.

2 Source inventory

Source	Type	Topics used	Exam relevance	Extraction notes
`MICRO_MANIFEST.md`	Manifest	Module 6 scope: MLRP/FOA, cost-minimizing compensation, DRM, envelope, virtual value, revenue equivalence; global four-layer frame and exercise requirements	Governs structure and notation	Text readable
`SKILL.md`	Builder protocol	Source inventory, pattern analysis, blueprint, packet structure, TeX rules, exercise distribution	Governs workflow	Text readable
`02ppl_agent.pdf`	Current problem set	Two moral-hazard exercises: continuum action/outcomes; two-output log utility with wealth parameter; FOA validity; cost function; insurance interpretation	High: mirrors 2021 Q2 and 2019 Q3	Text readable
`03ic-auc-app.pdf`	Current problem set	FPA as Bayesian game; symmetric bidding; all-pay-like modified auction; revenue comparison; reserve price and entry fee; virtual value regularity	High: mirrors 2022/2024 auction and DRM questions	Text readable

Source	Type	Topics used	Exam relevance	Extraction notes
`01signaling.pdf`	Current problem set	Information-economics notation, zero-profit wage logic, screening/adverse-selection cross-reference	Medium: boundary with Modules 4–5	Text readable
`June 2019.pdf, Q3`	Past qual	Continuous-output moral hazard: set up PA problem; test MLRP and FOA; solve cost-minimizing contract; derive cost; sufficient statistic question	Core moral hazard anchor	Text readable
`August 2021.pdf, Q2`	Past qual	Two-output moral hazard with log utility and initial wealth; FOA validity; cost-minimizing compensation; poorer agent comparative statics	Core moral hazard anchor	Text readable
`June 2022.pdf, Q3`	Past qual	Entrepreneur/lender moral hazard, CARA utility, high effort implementation, repayments, value of monitoring	Core Grossman–Hart and monitoring anchor	Text readable
`Spring 2017/2023 ECN 714 Final, Q3/Q1`	Past final	Linear contract $s(y) = a+by$; agent payoff with risk and effort cost; observed vs hidden effort	High for quick closed-form drills	Image-based; text partly not parsed
`Spring 2015 ECN 714 Final, Q3`	Past final	Restaurant manager moral hazard, revenue states, monitoring surprise visits, implement high effort	High for two-action contract drill	Image-based; text partly not parsed
`June 2023.pdf, Q4`	Past qual	Monopolist mechanism with finite types; single crossing; compact menu; monotone selections; two-type optimum	Core mechanism-design proof anchor	Text readable
`June 2024.pdf, Q4`	Past qual	IPVM DRM definition; single-bidder revenue maximization; checking whether a utility function is implementable by a DRM	Core auction/-DRM anchor	Text readable

Source	Type	Topics used	Exam relevance	Extraction notes
`August 2024.pdf, Q1-Q3`	Past/final	Single-bidder DRM utility/allocation tests; risk-averse FPA bidding formula	High for envelope and auction computation	Image-based; text partly parsed
`GitHub: Bayesian Games/die030-pagent.tex`	Lecture notes	Principal-agent timing, state-space and parametrized formulations, FOA Lagrangian, likelihood ratio FOC, Grossman–Hart two-step	Professor framing and notation	Web/HTML readable
`GitHub: Mechanism Design/die050-ipvm-auc.tex`	Lecture notes	IPVM, SPA dominance, auction as Bayesian game, FPA and variants	Professor framing and notation	Web/HTML readable
`GitHub: Mechanism Design/die060, die062, die064, die066, die067, die072`	Lecture notes	Mechanism design, one-dimensional IC, applications, optimal auctions, nonregular cases, Myerson–Satterthwaite style links	Supplementary source map	Directory readable; not all raw equations extracted

3 Pattern analysis and detailed blueprint

3.1 Recurring exam archetypes

A1. Set up a hidden-action problem. The prompt gives output distribution $f(x \mid a)$, utility $u(w) - c(a)$, and reservation utility $\bar{u}$. Write PC and either global IC or FOA IC. The most common loss is omitting the agent's maximization over all actions.

A2. Check MLRP/FOA and derive the likelihood-ratio contract. The prompt asks whether MLRP holds and whether FOA is valid. You must distinguish MLRP from CDFC/concavity of distribution function in effort. MLRP gives monotonicity of wages; it does not by itself always justify replacing global IC with the FOC.

A3. Use Grossman–Hart two-step. For each target action a, minimize expected wage cost subject to PC and IC; then choose a to maximize expected output minus the cost $C(a)$. This is the dominant ASU moral-hazard computation.

A4. Compute a two-output contract. With $x \in \{x_\ell, x_h\}$, write promised utilities v_h, v_ℓ and solve two linear equations: PC and local/global IC. Convert back to wages using $w = u^{-1}(v)$ or $w = e^v - \xi$ for log utility.

A5. Linear contract closed form. With $s(y) = a + by$, agent payoff usually has the form $a + b\bar{y}(e) - \frac{1}{2}[b^2\sigma^2(e) + e^2]$. The IC FOC pins down b, PC pins down a, and the principal then chooses b or e depending on observability.

A6. Monitoring and informativeness. If a signal is conditionally independent or statistically irrelevant conditional on output, it should not enter the optimal contract. If it changes the likelihood ratio, it may enter.

A7. Define a DRM and test implementability. For one-dimensional private values, a candidate allocation $q(x)$ is implementable iff it is nondecreasing. A candidate utility $U(x)$ is implementable iff it is convex, absolutely continuous, and $U'(x) = q(x)$ a.e. with $q \in [0,1]$ nondecreasing.

A8. Revenue equivalence / Myerson. Derive expected payment from the envelope formula, integrate by parts, define virtual value $\phi(x) = x - (1 - F(x))/f(x)$, and allocate iff virtual value is positive among winners under regularity.

A9. FPA bidding. Write a deviation as a fake type z or x and maximize $(v - b(z))F(z)^{n-1}$. The envelope method is faster than solving a differential equation from scratch.

3.2 Module map

Subtopic	Why it matters	Exam appearances / prototype	Core technique	Common trap
Hidden-action setup	Most PA questions award setup points before algebra	2019 Q3, 2021 Q2, 2022 Q3, Spring 2017/2023	PC + global IC, then optional FOA	Writing only the agent FOC as if sufficient
FOA and MLRP	Determines whether likelihood ratio condition is legitimate	2019 Q3, PS2	Lagrange multiplier on f_a/f; CDFC for validity	Confusing monotone wage with FOA validity
Grossman–Hart cost $C(a)$	Converts contract problem into single-variable action problem	2019 Q3, 2021 Q2, PS2	Utility promises; solve PC/IC; invert u	Minimizing utility instead of money cost
Two-output log contract	High-yield closed form	2021 Q2, PS2	Two linear equations in v_h, v_ℓ	Forgetting initial wealth ξ in wages
Linear share contract	Fast exam computation	Spring 2017/2023	Agent IC pins b; PC pins a; compare hidden vs observed effort	Choosing a before PC binds
Monitoring/informativeness	Tests use of existing theorem	2019 Q3(e), 2022 Q3(c), Spring 2015	Conditional likelihood ratio / sufficient statistic	Paying for signal that adds no incentive information
DRM definitions	Easy points, but definitions must be exact	2024 Q4, August 2024 Q1–Q2	Mechanism (q,t) plus truth-telling IC	Calling any menu a DRM without type reports
One-dimensional IC	Backbone of mechanism design	2023 Q4, 2024 Q4	Monotonicity + envelope	Missing absolute continuity / a.e. derivative

Subtopic	Why it matters	Exam appearances / prototype	Core technique	Common trap
Myerson auction	Repeated in auction problems	PS3, 2024 Q4	Virtual value and reserve cutoff	Maximizing surplus instead of virtual surplus
FPA and revenue equivalence	Standard auction computation	2022/2024 auction questions, PS3	Deviation-as-type, envelope, payment identity	Failing off-equilibrium bid check

3.3 Blueprint: canonical models and study sequence

1. Moral hazard environment and benchmark. Define timing, contract, PC, IC. Compare observable and hidden effort.
2. FOA and likelihood-ratio characterization. Derive the FOC for $w(x)$ under the relaxed problem. State MLRP, FOSD, and CDFC roles.
3. Grossman–Hart two-step. Derive $C(a)$ in utility terms and show how the principal chooses a.
4. Two-output log example. Solve PC and IC explicitly and prove poorer agents are weakly preferred by the principal.
5. Linear contract example. Compute a, b when effort is hidden and when observed.
6. Monitoring and sufficient statistic. State and use the likelihood-ratio test for whether a signal enters the contract.
7. Direct revelation mechanisms. Define mechanisms, BIC, IR, and revelation principle usage.
8. One-dimensional IC and envelope. Prove monotonicity and envelope; convert between q, U, t.
9. Myerson and auctions. Derive revenue formula, optimal allocation, reserve price, revenue equivalence, and FPA bidding formulas.
10. Exam practice. Complete exercise bank and past-qual map.

4 Learning objectives

After this module, you should be able to:

1. Given a moral-hazard environment, write the principal's problem with PC and global IC, then state what changes under the first-order approach.
2. Derive the likelihood-ratio contract equation and explain exactly which assumptions imply wage monotonicity and which assumptions justify FOA.
3. For a target effort/action, solve the Grossman–Hart cost-minimization problem in utility terms and convert promised utilities into wages or repayments.
4. Use a two-step cost function $C(a)$ to write the principal's outer problem and characterize the optimal action with KKT conditions when appropriate.
5. Identify whether a signal should enter compensation using likelihood ratios and the sufficient-statistic logic.
6. Define a direct revelation mechanism, BIC, interim utility, and IR in private-values environments.
7. Prove the one-dimensional IC envelope formula and monotonicity of the allocation rule.
8. Test whether a proposed utility or allocation function can be generated by a DRM.

9. Derive Myerson's virtual-value objective and the regular optimal auction rule.
10. Derive symmetric equilibrium bidding in SPA, FPA, and common variants under IPVM.

5 Canonical model 1: hidden action and the first-best benchmark

Visual intuition hook

Pair with Visual Intuition Curriculum: draw the game tree with contract offer, acceptance, unobserved effort, stochastic output, and transfers paid only after output.

5.1 Environment and timing

A risk-neutral principal hires a risk-averse agent. The agent chooses action $a \in A \subseteq \mathbb{R}_+$, unobserved by the principal. Output $x \in X \subseteq \mathbb{R}$ has density or probability mass $f(x \mid a)$ with full support. A contract is a measurable wage schedule $w : X \to \mathbb{R}$ or, equivalently, promised utility $v(x) = u(w(x))$. Agent utility is

$$U_A(w, a) = \mathbb{E}[u(w(x)) \mid a] - c(a),$$

where $u' > 0, u'' < 0, c' > 0, c'' \geq 0$. Principal payoff is

$$U_P(w, a) = \mathbb{E}[x - w(x) \mid a].$$

The timing is: principal offers $w(\cdot)$; agent accepts/rejects; if accepted, agent chooses a; output is realized; wage is paid.

5.2 Four-layer solution

Four-layer answer frame

Object. A contract $w(\cdot)$ and induced action a.
Feasibility. Participation and incentive compatibility.
Optimality/Equilibrium. Principal maximizes expected profit subject to those constraints.
Verification. Check PC, global IC, nonnegativity/domain, and whether any relaxed FOA solution is truly incentive compatible.

The hidden-action problem is

$$\begin{aligned}
\max_{a \in A,\, w(\cdot)} \quad & \int_X [x - w(x)] f(x \mid a)\, dx \\
\text{s.t.} \quad & \int_X u(w(x)) f(x \mid a)\, dx - c(a) \geq \bar{u}, && \text{(PC)} \\
& a \in \arg\max_{a' \in A} \left\{ \int_X u(w(x)) f(x \mid a')\, dx - c(a') \right\}. && \text{(IC)}
\end{aligned}$$

If effort were observable and contractible, IC is absent and the principal solves

$$\begin{aligned}
\max_{a, w(\cdot)} \quad & \int_X [x - w(x)] f(x \mid a)\, dx \\
\text{s.t.} \quad & \int_X u(w(x)) f(x \mid a)\, dx - c(a) \geq \bar{u}.
\end{aligned}$$

For a risk-neutral principal and risk-averse agent, the optimal full-information wage is constant across output realizations for a fixed a. The Lagrangian FOC in $w(x)$ gives

$$-f(x \mid a) + \lambda u'(w(x)) f(x \mid a) = 0,$$

so $u'(w(x)) = 1/\lambda$ for all x, hence $w(x)$ is constant. The hidden-action friction is exactly that a constant wage provides insurance but no marginal incentive.

Qualifier trap

Do not write "hidden action implies no first best" as a primitive. Prove it in the model: full insurance means constant wage; constant wage generally makes the agent choose the lowest-cost action unless preferences/technology make effort irrelevant or the agent is risk neutral.

6 Canonical model 2: FOA, likelihood ratios, MLRP, and wage monotonicity

Visual intuition hook

Pair with Visual Intuition Curriculum: draw output on the horizontal axis and the likelihood ratio $f_a(x \mid a)/f(x \mid a)$; the wage schedule inherits its slope from this statistic.

6.1 The relaxed first-order problem

Suppose A is an interval, densities are differentiable in a, and the agent's problem is differentiable. The first-order approach replaces global IC by

$$\int_X u(w(x)) f_a(x \mid a)\, dx = c'(a). \tag{FOA-IC}$$

The relaxed program is

$$\begin{aligned} \max_{a, w(\cdot)} \quad & \int_X [x - w(x)] f(x \mid a)\, dx \\ \text{s.t.} \quad & \int_X u(w(x)) f(x \mid a)\, dx - c(a) \geq \bar{u}, \\ & \int_X u(w(x)) f_a(x \mid a)\, dx = c'(a). \end{aligned}$$

The Lagrangian for fixed a is

$$\begin{aligned} \mathcal{L} = & \int_X [x - w(x)] f(x \mid a)\, dx + \lambda \left[\int_X u(w(x)) f(x \mid a)\, dx - c(a) - \bar{u} \right] \\ & + \mu \left[\int_X u(w(x)) f_a(x \mid a)\, dx - c'(a) \right]. \end{aligned}$$

The FOC with respect to $w(x)$ is

$$-f(x \mid a) + \lambda u'(w(x)) f(x \mid a) + \mu u'(w(x)) f_a(x \mid a) = 0.$$

Dividing by $u'(w(x)) f(x \mid a) > 0$ yields the central formula

$$\frac{1}{u'(w(x))} = \lambda + \mu \frac{f_a(x \mid a)}{f(x \mid a)}. \tag{LR}$$

For a risk-averse agent, $1/u'(w)$ is increasing in w. Thus if $\mu > 0$ and f_a/f is increasing in x, then $w(x)$ is increasing in x.

6.2 Definitions and theorem discipline

Definition 1 (MLRP)**.** *The family $f(\cdot \mid a)$ satisfies the monotone likelihood ratio property if, for $a' > a$, the likelihood ratio $f(x \mid a')/f(x \mid a)$ is increasing in x. Infinitesimally, this means $f_a(x \mid a)/f(x \mid a)$ is increasing in x.*

Definition 2 (CDFC)**.** *The distribution function $F(x \mid a)$ satisfies the convexity of distribution function condition if $F(x \mid a)$ is convex in a for each x. Depending on sign conventions, this is the condition used to make the agent's expected utility concave in action under increasing contracts.*

Proposition 1 (Exam version)**.** *MLRP plus the right concavity/CDFC conditions justify the first-order approach for increasing wage schedules. MLRP alone is typically enough to sign wage monotonicity from the likelihood-ratio FOC, but it is not by itself a complete proof that the agent's FOC is globally sufficient.*

6.3 Four-layer answer for an FOA question

Four-layer answer frame

Object. Candidate target action a and wage schedule $w(x)$.
Feasibility. PC and either global IC or FOA-IC plus conditions making FOA valid.
Optimality/Equilibrium. Solve the relaxed Lagrangian; derive (LR).
Verification. Check PC binds, FOA-IC holds, wage monotonicity if required, and global IC if FOA validity is not established.

7 Canonical model 3: Grossman–Hart cost-minimization approach

Visual intuition hook

Pair with Visual Intuition Curriculum: draw effort on the horizontal axis and two curves, expected output $\mathbb{E}[x \mid a]$ and implementation cost $C(a)$; the principal chooses the largest vertical gap.

Grossman–Hart's practical idea is to avoid jointly searching over actions and wage schedules. First compute the cheapest way to implement each action; then choose the best action.

7.1 Step 1: implementation cost

Let $v(x) = u(w(x))$ and let $h(v) = u^{-1}(v)$ be the money cost of delivering utility v. For each target action a, define

$$
\begin{aligned}
C(a) = \min_{v(\cdot)} \quad & \int_X h(v(x)) f(x \mid a)\, dx \\
\text{s.t.} \quad & \int_X v(x) f(x \mid a)\, dx - c(a) \geq \bar{u}, \\
& a \in \arg\max_{a' \in A} \left\{ \int_X v(x) f(x \mid a')\, dx - c(a') \right\}.
\end{aligned}
$$

Under FOA, replace global IC by

$$\int_X v(x) f_a(x \mid a)\, dx = c'(a).$$

The Lagrangian FOC for fixed a is

$$h'(v(x)) = \lambda + \mu \frac{f_a(x \mid a)}{f(x \mid a)}. \tag{GH}$$

Since $h'(v) = 1/u'(h(v))$, (GH) is the same likelihood-ratio equation.

7.2 Step 2: choose effort

Once $C(a)$ is known, the principal solves

$$\max_{a \in A} \mathbb{E}[x \mid a] - C(a).$$

If $A = [\underline{a}, \bar{a}]$ and the objective is differentiable, the KKT conditions are

$$\begin{aligned}
&\mathbb{E}_a[x \mid a^*] - C'(a^*) \le 0 && \text{if } a^* = \bar{a},\\
&\mathbb{E}_a[x \mid a^*] - C'(a^*) = 0 && \text{if } a^* \in (\underline{a}, \bar{a}),\\
&\mathbb{E}_a[x \mid a^*] - C'(a^*) \ge 0 && \text{if } a^* = \underline{a}.
\end{aligned}$$

Four-layer answer frame

Object. Cost function $C(a)$ and outer action a^*.
Feasibility. For each a, promised utility schedule satisfies PC and IC.
Optimality/Equilibrium. Minimize money cost for fixed a, then maximize expected output net of cost.
Verification. Confirm the fixed-a contract implements a and the outer action satisfies KKT/global comparison.

Qualifier trap

The inner problem minimizes expected *money* paid, not expected utility. Work in utility variables only because $h = u^{-1}$ converts promised utility into money cost.

8 Canonical model 4: two-output moral hazard with log utility

Visual intuition hook

Pair with Visual Intuition Curriculum: plot promised utility in the high-output state against promised utility in the low-output state; PC and IC are straight lines, and the least-cost point is their intersection for interior high effort.

This is the ASU 2021/PS2 workhorse. Output is $x \in \{x_\ell, x_h\}$, with $\Pr(x_h \mid a) = p(a)$, $p' > 0$, $p'' < 0$. Agent utility is

$$\log(\xi + w) - \psi(a),$$

where $\xi > 0$ is initial wealth and outside utility is $\log(\xi + \bar{w})$.

Let promised utilities be

$$v_h = \log(\xi + w_h), \qquad v_\ell = \log(\xi + w_\ell).$$

Then wages are $w_s = e^{v_s} - \xi$. For target a, the least-cost implementation problem is

$$\begin{aligned} \min_{v_h, v_\ell} \quad & p(a)(e^{v_h} - \xi) + (1 - p(a))(e^{v_\ell} - \xi) \\ \text{s.t.} \quad & p(a)v_h + (1 - p(a))v_\ell - \psi(a) \geq \log(\xi + \bar{w}), && \text{(PC)} \\ & a \in \arg\max_{a' \geq 0} \{p(a')v_h + (1 - p(a'))v_\ell - \psi(a')\}. && \text{(IC)} \end{aligned}$$

Under FOA, IC becomes

$$p'(a)(v_h - v_\ell) = \psi'(a). \tag{IC-FOC}$$

For $a > 0$, the cost-minimizing contract has PC and IC-FOC binding:

$$v_h - v_\ell = \frac{\psi'(a)}{p'(a)} \equiv \Delta(a),$$

$$p(a)v_h + (1 - p(a))v_\ell = \log(\xi + \bar{w}) + \psi(a) \equiv R(a).$$

Solving,

$$\begin{aligned} v_h(a) &= R(a) + (1 - p(a))\Delta(a), \\ v_\ell(a) &= R(a) - p(a)\Delta(a). \end{aligned}$$

Thus

$$\begin{aligned} w_h(a) &= \exp\{R(a) + (1 - p(a))\Delta(a)\} - \xi, \\ w_\ell(a) &= \exp\{R(a) - p(a)\Delta(a)\} - \xi. \end{aligned}$$

The cost function is

$$C(a; \xi) = p(a)e^{R(a) + (1 - p(a))\Delta(a)} + (1 - p(a))e^{R(a) - p(a)\Delta(a)} - \xi.$$

Since $e^{R(a)} = (\xi + \bar{w})e^{\psi(a)}$, this can be written

$$C(a; \xi) = (\xi + \bar{w})e^{\psi(a)} \left[p(a)e^{(1 - p(a))\Delta(a)} + (1 - p(a))e^{-p(a)\Delta(a)}\right] - \xi.$$

8.1 Why the principal weakly prefers a poorer agent

Differentiate $C(a; \xi)$ with respect to ξ:

$$\frac{\partial C}{\partial \xi} = e^{\psi(a)} \left[pe^{(1-p)\Delta} + (1 - p)e^{-p\Delta}\right] - 1.$$

The bracket is at least 1 by Jensen because it is the expectation of e^Z where Z takes values $(1-p)\Delta$ and $-p\Delta$ with mean zero. Therefore $\partial C / \partial \xi \geq e^{\psi(a)} - 1 \geq 0$. For positive effort, the inequality is strict under the usual assumptions. A wealthier agent has lower marginal utility variation, so inducing effort requires a larger money spread.

Four-layer answer frame

Object. Promised utilities (v_h, v_ℓ) for each target a.
Feasibility. PC and IC; under valid FOA, $p'(a)(v_h - v_\ell) = \psi'(a)$.
Optimality/Equilibrium. PC and IC bind; solve two linear equations; convert to wages.
Verification. Check $v_h \geq v_\ell$, domain $\xi + w_s > 0$, FOA/global IC, and outer maximization.

9 Canonical model 5: linear contracts and observed effort

Visual intuition hook

Pair with Visual Intuition Curriculum: draw the agent's expected utility as a concave parabola in effort; a higher share b rotates the marginal-benefit line upward.

The Spring 2017/2023 style problem gives output y with mean $\bar{y}(e)$ and variance $\sigma^2(e)$, and restricts contracts to $s(y) = a + by$. The agent's expected utility is

$$U^A(e; a, b) = a + b\bar{y}(e) - \frac{1}{2}\left[b^2\sigma^2(e) + e^2\right].$$

If $\bar{y}(e) = e$ and $\sigma^2(e) = \sigma^2$ is constant, then the hidden-effort IC FOC is

$$\frac{\partial U^A}{\partial e} = b - e = 0 \quad \Rightarrow \quad e = b.$$

PC binds at optimum:

$$a + b^2 - \frac{1}{2}(b^2\sigma^2 + b^2) = 0,$$

so

$$a = \frac{1}{2}b^2\sigma^2 - \frac{1}{2}b^2.$$

The principal's expected payoff is

$$\Pi(b) = \bar{y}(b) - a - b\bar{y}(b) = b - a - b^2 = b - \frac{1}{2}b^2(1 + \sigma^2).$$

The optimal share is

$$b^* = \frac{1}{1 + \sigma^2}, \qquad e^* = \frac{1}{1 + \sigma^2},$$

and a^* follows from PC.

If effort is observable, the principal may choose $(a(e), b(e), e)$ directly. Since incentives are unnecessary, set $b = 0$ to avoid risk exposure, bind PC with $a = e^2/2$, and choose

$$\max_e \ e - \frac{1}{2}e^2,$$

so $e^* = 1$ if unconstrained. This is the first-best benchmark: full insurance and efficient effort.

Four-layer answer frame

Object. Linear contract parameters (a, b) and induced effort e.
Feasibility. PC; hidden effort also requires agent IC.
Optimality/Equilibrium. IC pins effort as a function of b; PC pins a; principal chooses b.
Verification. Check second-order condition in agent effort, PC binding, and any constraints on b or e.

10 Canonical model 6: monitoring, informativeness, and sufficient statistics

Visual intuition hook

Pair with Visual Intuition Curriculum: draw two signals with identical likelihood-ratio contours; if adding y does not change the contour relevant for effort, it should not change compensation.

Suppose output x and signal y have joint density $q(x, y \mid a)$. Under FOA the wage FOC is

$$\frac{1}{u'(w(x,y))} = \lambda + \mu \frac{q_a(x, y \mid a)}{q(x, y \mid a)}.$$

Therefore y enters the optimal contract if and only if it changes the likelihood ratio relevant for effort, after conditioning on what is already observed.

Proposition 2 (Sufficient-statistic test, exam form)**.** *If*

$$\frac{q_a(x, y \mid a)}{q(x, y \mid a)} = r(x, a)$$

is independent of y, then an optimal contract need not condition on y. If the likelihood ratio varies with y on a set of positive probability and the IC multiplier is nonzero, conditioning on y can improve incentives or risk sharing.

Proof. The FOC shows that promised utility depends on (x, y) only through the likelihood ratio. If the likelihood ratio is a function of x alone, any variation of wages with y adds risk without relaxing PC or IC. Jensen then improves or weakly preserves cost by replacing $w(x, y)$ with a certainty-equivalent conditional on x. □

11 Canonical model 7: direct revelation mechanisms

Visual intuition hook

Pair with Visual Intuition Curriculum: draw reports as arrows from types to contracts; IC says every type's arrow points to its own assigned contract.

Consider agents with types $\theta_i \in \Theta_i$. A direct revelation mechanism asks each agent to report a type and maps reported types into allocations and transfers.

For one object and quasilinear private values, a DRM is

$$(q_i(\hat{\theta}), t_i(\hat{\theta}))_{i=1}^{I},$$

where $q_i(\hat{\theta}) \in [0, 1]$, $\sum_i q_i(\hat{\theta}) \leq 1$, and $t_i(\hat{\theta})$ is the payment by agent i to the designer. Agent i's utility is

$$\theta_i q_i(\hat{\theta}) - t_i(\hat{\theta}).$$

Truthful reporting is Bayesian incentive compatible if for every i, every type θ_i, and every misreport $\hat{\theta}_i$,

$$\mathbb{E}_{\theta_{-i}}[\theta_i q_i(\theta_i, \theta_{-i}) - t_i(\theta_i, \theta_{-i}) \mid \theta_i] \geq \mathbb{E}_{\theta_{-i}}[\theta_i q_i(\hat{\theta}_i, \theta_{-i}) - t_i(\hat{\theta}_i, \theta_{-i}) \mid \theta_i].$$

Four-layer answer frame

Object. Allocation rule q and transfer rule t.
Feasibility. Resource feasibility, BIC, and IR.
Optimality/Equilibrium. Designer maximizes expected objective over direct truthful mechanisms.
Verification. Check monotonicity/envelope conditions or direct IC inequalities, then IR and feasibility.

12 Canonical model 8: one-dimensional IC, envelope, and implementability tests

Visual intuition hook

Pair with Visual Intuition Curriculum: graph interim utility $U(x)$; its slope is allocation probability $Q(x)$, so implementability means utility is convex and its slope lies between 0 and 1.

Let a single bidder have type $x \in [\underline{x}, \bar{x}]$ and utility $xq(x) - t(x)$. Define interim utility

$$U(x) = xq(x) - t(x).$$

For a direct mechanism, IC means

$$\begin{aligned} U(x) &\geq xq(z) - t(z) = U(z) + (x - z)q(z) \quad \forall x, z, \\ U(z) &\geq zq(x) - t(x) = U(x) + (z - x)q(x) \quad \forall x, z. \end{aligned}$$

Adding and rearranging for $x > z$ gives

$$q(x) \geq q(z).$$

Thus IC implies monotonicity of q. The same inequalities imply

$$q(z) \leq \frac{U(x) - U(z)}{x - z} \leq q(x),$$

so, where differentiable,

$$U'(x) = q(x). \tag{Envelope}$$

Hence

$$U(x) = U(\underline{x}) + \int_{\underline{x}}^{x} q(s)\, ds. \tag{Utility formula}$$

Transfers are recovered by

$$t(x) = xq(x) - U(x). \tag{Payment formula}$$

Proposition 3 (One-dimensional implementability)**.** *An allocation rule $q : [\underline{x}, \bar{x}] \to [0, 1]$ is implementable iff it is nondecreasing. Given nondecreasing q, any constant $U(\underline{x})$ defines an IC transfer rule by the utility and payment formulas. IR usually pins $U(\underline{x}) \geq 0$, with equality at revenue optimum.*

12.1 How to test a proposed utility function

A proposed U for a single-bidder DRM is implementable only if:

1. U is convex, because it is the supremum of affine functions $xq(z) - t(z)$;
2. U is absolutely continuous on intervals used in the problem;
3. $U'(x)$ exists a.e. and lies in $[0, 1]$;
4. $U'(x)$ is nondecreasing a.e.;
5. IR holds if required, usually $U(x) \geq 0$.

When those conditions hold, set $q(x) = U'(x)$ a.e. and $t(x) = xq(x) - U(x)$, with arbitrary monotone-consistent definitions at kinks.

Qualifier trap

A utility function can be increasing but fail implementability because its derivative decreases somewhere. In one-dimensional quasilinear mechanisms, the allocation is the slope of utility.

13 Canonical model 9: Myerson optimal auction

Visual intuition hook

Pair with Visual Intuition Curriculum: draw virtual value $\phi(x)$; the reserve price is where the graph crosses zero, and the object goes to the highest nonnegative virtual value.

Consider I risk-neutral bidders with independent private values $x_i \sim F_i$ with densities $f_i > 0$. A direct mechanism has interim allocation $Q_i(x_i) = \mathbb{E}_{x_{-i}} q_i(x_i, x_{-i})$ and interim payment $T_i(x_i) = \mathbb{E}_{x_{-i}} t_i(x_i, x_{-i})$. IC implies

$$U_i(x_i) = U_i(\underline{x}_i) + \int_{\underline{x}_i}^{x_i} Q_i(s)\, ds,$$

and

$$T_i(x_i) = x_i Q_i(x_i) - U_i(x_i).$$

Expected payment is

$$\begin{aligned}
\mathbb{E}[T_i(x_i)] &= \int_{\underline{x}_i}^{\bar{x}_i} \left[x_i Q_i(x_i) - U_i(\underline{x}_i) - \int_{\underline{x}_i}^{x_i} Q_i(s)\, ds \right] f_i(x_i)\, dx_i \\
&= \int_{\underline{x}_i}^{\bar{x}_i} \left[x_i - \frac{1 - F_i(x_i)}{f_i(x_i)} \right] Q_i(x_i) f_i(x_i)\, dx_i - U_i(\underline{x}_i).
\end{aligned}$$

Define virtual value

$$\phi_i(x) = x - \frac{1 - F_i(x)}{f_i(x)}.$$

Expected revenue equals expected virtual surplus minus information rents at the bottom:

$$\mathbb{E}\left[\sum_i t_i(x) \right] = \mathbb{E}\left[\sum_i \phi_i(x_i) q_i(x) \right] - \sum_i U_i(\underline{x}_i).$$

At the revenue optimum, set $U_i(\underline{x}_i) = 0$ if IR binds at the lowest type. Under regularity, ϕ_i increasing, choose feasible q to maximize pointwise virtual surplus:

$$q_i(x) = 1 \quad \text{only if} \quad \phi_i(x_i) = \max_j \phi_j(x_j) \geq 0.$$

In the symmetric regular case, this is a second-price auction with reserve r solving

$$\phi(r) = r - \frac{1 - F(r)}{f(r)} = 0.$$

Four-layer answer frame

Object. Allocation rule $q(x)$ and payments $t(x)$.
Feasibility. Feasible allocation, IC monotonicity, IR.
Optimality/Equilibrium. Maximize virtual surplus pointwise under regularity.
Verification. Check monotonicity of allocation, reserve cutoff, payment formula, and bottom-type rent.

14 Canonical model 10: revenue equivalence and first-price bidding

Visual intuition hook

Pair with Visual Intuition Curriculum: draw the bidder shading curve below the 45-degree truthful-bid line; shading is the expected information rent from raising type.

In any symmetric IPVM auction with the same allocation probability $Q(v)$ and the same utility for the lowest type, the interim utility is

$$U(v) = U(0) + \int_0^v Q(s)\,ds.$$

If $U(0) = 0$, expected payments are pinned down. This is the revenue equivalence logic.

14.1 Second-price auction

Truthful bidding $b(v) = v$ is weakly dominant because a bidder's bid determines whether she wins but not the price paid conditional on winning, except at ties. If the highest rival bid is m, then winning is profitable iff $v \geq m$.

14.2 First-price auction: risk-neutral symmetric equilibrium

With n bidders and i.i.d. F on $[0, \bar{v}]$, suppose b is strictly increasing. A type v who bids as if type z obtains

$$(v - b(z))F(z)^{n-1}.$$

At equilibrium $z = v$. The envelope approach is faster. The allocation probability is

$$Q(v) = F(v)^{n-1}.$$

With $U(0) = 0$,

$$U(v) = \int_0^v F(s)^{n-1}\,ds.$$

But in a first-price auction,

$$U(v) = (v - b(v))F(v)^{n-1}.$$

Therefore

$$b(v) = v - \frac{\int_0^v F(s)^{n-1}\,ds}{F(v)^{n-1}}. \tag{FPA}$$

For uniform $[0,1]$, $F(v) = v$, and

$$b(v) = v - \frac{v^n/n}{v^{n-1}} = \frac{n-1}{n}v.$$

For two bidders uniform, $b(v) = v/2$.

14.3 Risk-averse FPA with utility $(v-b)^\alpha$

If utility from winning is $(v-b)^\alpha$ with $\alpha \geq 1$, a symmetric increasing equilibrium satisfies

$$U(v) = (v - b_\alpha(v))^\alpha F(v)^{n-1}.$$

The envelope condition gives

$$U'(v) = \alpha(v - b_\alpha(v))^{\alpha-1} F(v)^{n-1}.$$

Solving the standard differential equation yields the ASU-style formula

$$b_\alpha(v) = v - \int_0^v \left(\frac{F(x)}{F(v)}\right)^{\alpha(n-1)} dx. \tag{RA-FPA}$$

When $\alpha = 1$, this reduces to the risk-neutral first-price formula.

15 Canonical model 11: reserve price and entry fee

Visual intuition hook

Pair with Visual Intuition Curriculum: draw bidder surplus $U(v)$; an entry fee subtracts a rectangle from every participant, while a reserve truncates low types from allocation.

In a regular symmetric IPVM, Myerson's reserve maximizes virtual surplus. The optimal reserve r^* solves

$$r^* - \frac{1-F(r^*)}{f(r^*)} = 0$$

when the solution is interior. An entry fee can extract ex ante surplus from participants, but in a standard auction with voluntary participation before type realization, the maximum fee equals ex ante expected bidder surplus from entering. If participation occurs after observing type, the fee acts like a reserve because low types decline to participate.

Qualifier trap

A reserve price changes the allocation rule; an entry fee changes participation and rents. They are not mechanically interchangeable unless the timing makes them equivalent.

16 Core derivation templates

Reusable template

Hidden-action setup template.

1. Write timing and contractible variables.
2. Principal's objective: expected output minus expected payments.
3. PC: agent expected utility from accepting is at least outside option.
4. IC: target action maximizes agent expected utility over all actions.
5. Only after that, state FOA and its required validity assumptions.

Reusable template

Grossman–Hart template. For each a: define $v = u(w)$ and $h = u^{-1}$. Minimize $\mathbb{E}[h(v(x)) \mid a]$ subject to PC and IC. Derive FOC $h'(v) = \lambda + \mu f_a/f$. Set $C(a)$ equal to minimized cost. Then solve $\max_a \mathbb{E}[x \mid a] - C(a)$.

Reusable template

One-dimensional mechanism template. IC inequalities imply monotone allocation. Envelope gives $U' = Q$. Integrate to get $U(x) = U(\underline{x}) + \int_{\underline{x}}^{x} Q(s)ds$. Recover payment by $T = xQ - U$. Use IR to pin the constant.

Reusable template

Myerson template. Use envelope to replace payments with virtual surplus. Define $\phi = x - (1-F)/f$. Under regularity, allocate to the highest nonnegative virtual value. Recover payments by critical values or payment identity.

17 Exam variations

Exam/source	Variation	What stays invariant	Fast response
2019 Q3	Continuous output density $f(x \mid a) = 1 + .5(1-2x)(1-2a)$; asks MLRP, FOA, cost-min contract, signal y	PC/IC setup; likelihood-ratio FOC; Grossman–Hart; sufficient statistic	Compute f_a/f, sign monotonicity, then derive $v(x)$ from $h'(v) = \lambda + \mu f_a/f$
2021 Q2 / PS2	Two outputs, log utility, initial wealth ξ	PC and IC solve linearly in promised utilities	Solve $v_h - v_l = \psi'/p'$ and weighted average PC

Exam/source	Variation	What stays invariant	Fast response
2022 Q3	Lender/entrepreneur with CARA utility and repayments	Implement high effort with PC and IC; value of observing effort is first-best minus moral-hazard cost	Work in income utility or certainty equivalents
Spring 2017/2023	Linear contract $a+by$ with mean effort and variance risk	Agent IC pins b; PC pins a; observed effort removes incentive wedge	Derive $b = 1/(1+\sigma^2)$ in constant-variance case
Spring 2015	Restaurant revenue states; monitoring surprise visits	Two-action IC and PC; monitoring creates additional contractible state	Write wages by revenue/-monitoring event; compare implementation costs
2023 Q4	Finite-type monopolist mechanism, single crossing, monotone selections	IC inequalities imply monotone allocations/transfers; low IR and high IC bind in two-type optimum	Use Module 5 binding-constraint logic plus mechanism language
2024 Q4 / August 2024 Q1–Q2	DRM definition; utility/allocation implementability tests	Allocation monotone; utility convex with slope allocation	Check U' in $[0,1]$ and nondecreasing; recover q, t
PS3 / 2024 auctions	FPA, modified auction, reserve/entry fee, risk aversion	Deviation-as-type; envelope; revenue equivalence	Write $Q(v)$, integrate utility, recover bid/payment

18 Common traps

1. **Skipping global IC.** FOA is a theorem-dependent shortcut, not the definition of incentive compatibility.
2. **Using MLRP as FOA validity.** MLRP helps monotone wages; FOA validity also requires concavity/single-crossing-type conditions.
3. **Forgetting PC binds.** With risk-neutral principal and transferable money, PC usually binds unless limited liability or nonconvexities obstruct it.
4. **Confusing utility wages with money wages.** In log utility, $v = \log(\xi + w)$ and $w = e^v - \xi$.
5. **Wrong direction of monotonicity.** Higher output gets higher wage if the likelihood ratio is increasing and the IC multiplier is positive.
6. **No verification after solving two equations.** The two equations solve a candidate; global IC, domain, and action bounds still need checking.
7. **Calling a menu a DRM without reports.** A direct revelation mechanism maps reported types to outcomes.

8. **Ignoring feasibility in auctions.** Allocation probabilities must sum to at most one state by state.
9. **Utility implementability without convexity.** A bidder utility must be convex because it is the upper envelope of affine functions.
10. **Using surplus instead of virtual surplus.** Revenue maximization allocates by virtual values, not values, unless virtual values preserve the same ordering and reserve is zero.
11. **Misplacing the entry fee.** Whether it is paid before or after type realization changes the participation constraint.
12. **Forgetting critical payments.** Once allocation is chosen, payments must make truthful reporting optimal; they are not arbitrary.

19 Exercise bank

19.1 Mechanical drills

M1. Skill: write hidden-action constraints. Let $A = \{0, 1\}$ and outputs $x \in \{0, 10\}$ with $p_1 > p_0$ probability of high output. Write PC and IC to implement $a = 1$.

M2. Skill: FOA likelihood ratio. For density $f(x \mid a) = 1 + .5(1-2x)(1-2a)$ on $[0, 1]$, compute f_a/f.

M3. Skill: two-output promised utilities. Given $p(a) = a/(1 + a)$ and $\psi(a) = a^2/2$, write $v_h(a), v_l(a)$ under the log-utility two-output formula.

M4. Skill: linear contract. With $\bar{y}(e) = e$ and constant variance $\sigma^2 = 3$, compute b^* and induced effort.

M5. Skill: DRM definition. Write a feasible DRM for one bidder with value $x \in [0, 1]$ and allocation $q(x) = \mathbf{1}\{x \geq r\}$.

M6. Skill: payment recovery. Given $q(x) = x$ on $[0, 1]$ and $U(0) = 0$, compute $U(x)$ and $t(x)$.

19.2 Proof and characterization drills

P1. Skill: monotone wages. Prove that if f_a/f is increasing in output and $\mu > 0$, then the FOA wage schedule is increasing.

P2. Skill: one-dimensional IC. Prove that IC implies monotone allocation and the envelope inequalities $q(z) \leq [U(x) - U(z)]/(x - z) \leq q(x)$ for $x > z$.

P3. Skill: utility implementability. Determine whether $U(x) = 2x^2/3$ on $[0, 1]$ can be generated by a single-bidder DRM.

P4. Skill: virtual value. Derive expected revenue $\mathbb{E}[t] = \mathbb{E}[\phi(x)q(x)] - U(0)$ for one bidder.

P5. Skill: sufficient statistic. Prove that a signal y is not used when $q_a(x, y \mid a)/q(x, y \mid a)$ is independent of y.

P6. Skill: poorer agent comparative statics. In the two-output log model, prove $\partial C(a; \xi)/\partial \xi \geq 0$.

19.3 Past-qual-style applications

Q1. 2019 Q3 style. Set up the continuous-output PA problem, test MLRP, and state whether FOA is valid. Then write the cost-minimization problem for each a.

Q2. 2021 Q2 style. Solve the two-output log-utility contract for a fixed effort and state the principal's outer problem.

Q3. 2022 Q3 style. A lender wants to implement high effort for a CARA entrepreneur with two returns. Write PC, IC, and the repayment minimization problem.

Q4. Spring 2017/2023 style. Solve for a, b in a linear contract when variance is constant; then compare to observable effort.

Q5. 2024 Q4 style. Let $F(x) = x^2$ for $x \in [0, 1]$. Find the single-bidder regular optimal reserve and allocation rule.

Q6. PS3 style. Derive the symmetric FPA bid for $n = 2$, uniform $[0, 1]$, and compare expected revenue to SPA.

19.4 Trap / economist-claim questions

T1. Claim: "MLRP means the first-order approach is valid." Evaluate.

T2. Claim: "Any increasing utility function $U(x)$ can be a bidder's IC utility." Evaluate.

T3. Claim: "The revenue-maximizing auction always allocates to the highest valuation." Evaluate.

T4. Claim: "If effort becomes observable, the agent must bear more risk." Evaluate.

T5. Claim: "An unused signal is always worthless." Evaluate.

20 Answer sketches

20.1 Mechanical drills

M1. PC: $p_1 u(w_H) + (1-p_1)u(w_L) - c(1) \geq \bar{u}$. IC: $p_1 u(w_H) + (1-p_1)u(w_L) - c(1) \geq p_0 u(w_H) + (1-p_0)u(w_L) - c(0)$. Rearranged: $(p_1 - p_0)[u(w_H) - u(w_L)] \geq c(1) - c(0)$.

M2. $f_a(x \mid a) = -(1-2x) = 2x - 1$. Thus $f_a/f = (2x-1)/[1 + .5(1-2x)(1-2a)]$. Check monotonicity by differentiating; it is increasing for admissible a.

M3. $p' = 1/(1+a)^2$, so $\Delta(a) = \psi'(a)/p'(a) = a(1+a)^2$. $R(a) = \log(\xi + \bar{w}) + a^2/2$. Use $v_h = R + (1-p)\Delta$ and $v_l = R - p\Delta$.

M4. $b^* = 1/(1+\sigma^2) = 1/4$, so induced effort is $e^* = 1/4$.

M5. Set $q(x) = \mathbf{1}\{x \geq r\}$ and $t(x) = r\mathbf{1}\{x \geq r\}$. This is a posted price; IC follows because types above r buy and types below do not.

M6. $U(x) = \int_0^x s\, ds = x^2/2$. Payment $t(x) = xq(x) - U(x) = x^2 - x^2/2 = x^2/2$.

20.2 Proof and characterization drills

P1. From $1/u'(w(x)) = \lambda + \mu f_a/f$. Since $u'' < 0$, $1/u'(w)$ is strictly increasing in w. If f_a/f is increasing and $\mu > 0$, the right side is increasing in x, so $w(x)$ is increasing.

P2. IC gives $U(x) \geq U(z) + (x-z)q(z)$ and $U(z) \geq U(x) + (z-x)q(x)$. Rearrange. The two inequalities imply $q(z) \leq q(x)$ and the slope bounds.

P3. $U'(x) = 4x/3$. This exceeds 1 for $x > 3/4$, so it cannot be an allocation probability. Not implementable on $[0,1]$.

P4. Substitute $t = xq - U$, $U = U(0) + \int_0^x q(s)ds$, integrate, and swap integration order: $\int_0^1 \int_0^x q(s)ds f(x)dx = \int_0^1 q(s)[1-F(s)]ds$.

P5. If the likelihood ratio is independent of y, the FOC gives promised utility independent of y except for risk-sharing noise. Conditional Jensen lowers expected wage cost by replacing y-contingent wages with a conditional certainty equivalent.

P6. Use the cost formula. The derivative equals $e^{\psi(a)}\mathbb{E}[e^Z] - 1$ where $\mathbb{E}Z = 0$. Jensen gives $\mathbb{E}e^Z \geq 1$, so the derivative is nonnegative.

20.3 Past-qual-style applications

Q1. Write the full PA program. Compute f_a/f. Establish MLRP by monotonicity. For FOA, cite the additional concavity/CDFC condition; if not verified, label FOA as a relaxed candidate and verify global IC separately.

Q2. Use $v_h - v_l = \psi'/p'$ and PC. Convert to wages $w_s = e^{v_s} - \xi$. Principal maximizes $p(a)x_h + (1-p(a))x_l - C(a)$.

Q3. Let incomes be $I_H = \bar{\theta} - \bar{x}$ and $I_L = \theta - x$. PC uses expected CARA utility under $e = 1$. IC compares expected utility under $e = 1$ to $e = 0$. Minimize expected repayments subject to lender break-even/investment feasibility as specified.

Q4. Hidden effort: $e = b$, PC pins a, principal chooses $b = 1/(1+\sigma^2)$. Observable effort: set $b = 0$, PC pins fixed wage, choose efficient effort $e = 1$ if unconstrained.

Q5. $F = x^2$, $f = 2x$, so $\phi(x) = x - (1-x^2)/(2x) = (3x^2-1)/(2x)$. Reserve solves $3r^2 - 1 = 0$, hence $r = 1/\sqrt{3}$. Allocate iff $x \geq r$.

Q6. FPA bid is $v/2$. SPA truthful bids generate expected revenue equal to expected second order statistic, $1/3$. FPA revenue is expected winning bid $\mathbb{E}[\max(v_1, v_2)/2] = 1/3$.

20.4 Trap questions

T1. False. MLRP signs the wage schedule. FOA validity requires the agent's objective to be concave or otherwise globally single-peaked in effort.

T2. False. IC utility must be convex with nondecreasing derivative equal to an allocation probability in $[0, 1]$.

T3. False. Under revenue maximization, allocation follows virtual values. If virtual values are regular and symmetric, this is highest value above reserve; otherwise ironing or asymmetry can alter allocation.

T4. False. Observable effort removes the incentive need for risk exposure. With risk-neutral principal and risk-averse agent, full insurance is optimal for fixed observable effort.

T5. False. A signal may be unused at the optimum because it adds no information conditional on output, but it may still have value in other environments, under limited liability, or if the available output measure changes.

21 Past-qual practice map

Year/source	Problem	Module skill	Priority
June 2019 Q3	Continuous-output moral hazard; MLRP/FOA; cost-min contract; signal use	Full PA setup, FOA, GH, sufficient statistic	Very high
August 2021 Q2	Two-output log moral hazard; poorer agent	Utility promises, cost function, comparative statics	Very high
June 2022 Q3	Entrepreneur/lender moral hazard; CARA; monitoring value	High effort implementation, repayment design, value of observability	Very high
Spring 2017 ECN 714 Q3	Linear contract $a + by$, hidden vs observed effort	Closed-form share contract	High
Spring 2023 ECN 714 Q1	Same linear-contract structure with variance dependence	Candidate contract and observability	High
Spring 2015 ECN 714 Q3	Restaurant manager; monitoring	Two-action IC, monitoring states	High
June 2023 Q4	Finite-type monopolist mechanism	Single crossing, monotone menus, two-type binding constraints	High
June 2024 Q4	DRM and single-bidder optimal mechanism	Definition, Myerson, utility implementability	Very high
August 2024 Q1–Q2	Utility/allocation implementability for DRM	Envelope tests, monotonicity	Very high
August 2024 Q3	Risk-averse first-price auction	FPA bidding derivation	High
Problem set 2	Moral hazard models	Current-course anchor	Very high

Year/source	Problem	Module skill	Priority
Problem set 3	IPVM auctions and variants	Current-course anchor	Very high

22 Final study checklist

Before attempting old quals on this module, verify that you can do each item without notes:

1. Write a hidden-action principal-agent problem with PC and global IC.
2. State the FOA constraint and derive the likelihood-ratio FOC.
3. Explain MLRP, CDFC, and why they are not the same condition.
4. Solve a two-output contract in promised utilities and convert to wages.
5. Derive the Grossman–Hart cost function $C(a)$ and outer problem.
6. Solve the linear contract model with $s(y) = a + by$.
7. Use a likelihood-ratio argument to decide whether a signal enters compensation.
8. Define a DRM, BIC, and IR precisely.
9. Prove IC implies allocation monotonicity and the envelope formula.
10. Test a proposed $U(x)$ for implementability by checking convexity and slope.
11. Derive the payment formula from q and $U(\underline{x})$.
12. Derive virtual value and Myerson's regular optimal auction rule.
13. Derive FPA bidding from revenue equivalence/envelope.
14. Explain reserve prices and entry fees without confusing their timing.
15. End every solution with verification: feasibility, IC/IR, optimality, and boundary cases.

23 Quality audit

- Source inventory included, with accessible sources and extraction issues flagged.
- Pattern analysis and blueprint precede the full packet.
- Each canonical model uses the four-layer answer frame.
- Moral hazard derivations include hidden-action setup, FOA, likelihood ratios, Grossman–Hart, two-output log contracts, linear contracts, and monitoring.
- Mechanism-design derivations include DRM, BIC/IR, monotonicity, envelope, payment recovery, Myerson, revenue equivalence, and FPA bidding.
- Exercises follow the requested distribution: mechanical, proof/characterization, past-qual applications, and trap/economist-claim questions.
- Remaining limitation: some old ECN 714 finals are image-based, so their content is incorporated from the visible page text rather than fully parsed source text.